
Uncovering Conceptual Blindspots in Generative Image Models Using Sparse Autoencoders

Anonymous Author(s)

Affiliation

Address

email

Abstract

Although generative image models trained on large-scale corpora now exhibit striking empirical competence, they continue to fail, with surprising regularity, on ostensibly elementary concepts—e.g., human hands or objects appearing in groups of four—that one would reasonably expect to be amply represented in their training distributions. Because these failures have primarily been reported through anecdotal observation, it remains unclear whether they constitute isolated irregularities attached to a small set of unusually difficult concepts or instead reveal more systematic constraints in the representational and generative capacities of contemporary models. To investigate this issue, we introduce a systematic methodology for identifying and characterizing "conceptual blindspots"—concepts that are present in the training data but absent, attenuated, or otherwise misrepresented in a model's generations. Our method uses sparse autoencoders (SAEs) to derive interpretable concept embeddings, thereby enabling a quantitative comparison between the prevalence of concepts in real images and their prevalence in generated counterparts. We train an archetypal SAE (RA-SAE) on DINOv2 features with 32,000 concepts—the largest such SAE to date—which supports fine-grained measurement of conceptual disparities. When applied to four widely used generative models (Stable Diffusion 1.5/2.1, PixArt, and Kandinsky), the proposed analysis exposes both suppressed blindspots (e.g., bird feeders, DVD discs, and whitespaces on documents) and exaggerated blindspots (e.g., wood background texture and palm trees). At the individual-datapoint level, the same framework further isolates memorization artifacts, namely instances in which models reproduce highly specific visual templates encountered during training. Taken together, our results establish a theoretically grounded framework for systematically diagnosing conceptual blindspots in generative models by evaluating their conceptual fidelity relative to the underlying data-generating process.

1 Introduction

Generative image models trained on large-scale datasets have attained unprecedented capabilities, thereby enabling their deployment across a broad range of applications within computer vision [1–9] as well as in domains that extend substantially beyond conventional visual synthesis [9–15]. Yet, notwithstanding these advances, both qualitative [16–18] and quantitative investigations [19–20] indicate that such models can fail to render comparatively simple concepts, including human hands [21–24], objects appearing in groups of four [25], and negations or object relations [20, 26]. Indeed, when prompted to generate images containing these concepts, models often synthesize outputs that contain adjacent or superficially related structures while failing to instantiate the intended ground-truth concept, as in the familiar production of hands with six



Figure 1: $\delta(k)$ quantifies the extent to which a model over- or under-generates a concept c_k relative to its empirical frequency in natural data. We designate concepts with $\delta(k) < 0.1$ as **suppressed conceptual blindspots** and concepts with $\delta(k) > 0.9$ as **exaggerated conceptual blindspots**. The displayed images, produced by four widely used generative image models, illustrate both conceptually misaligned generations and comparatively aligned concepts.

fingers. Such failure modes, which we call “conceptual blindspots”¹ are counterintuitive precisely because the relevant concepts appear likely to have been observed repeatedly, and with substantial visual variation, during training. This tension raises a central question: do these failures merely reflect intriguing anomalies associated with a few idiosyncratic concepts, or do they instead instantiate a broader phenomenon whereby models systematically under-produce or over-produce images containing particular regions of conceptual space?

Existing evaluation methodologies for generative image models are poorly suited to answering this question at the required conceptual granularity [28, 29]. In particular, many standard protocols rely on aggregate measures intended primarily to assess perceptual realism, such as FID [30], and therefore obscure distributional failures that manifest only at the level of specific concepts. Metrics such as CLIPScore, generation diversity, or distribution-coverage statistics [31–33] provide partial evidence about model behavior, but because they do not operate at the resolution of fine-grained visual features or concepts [34, 35], they offer limited leverage for isolating conceptual blindspots. Similarly, qualitative evaluation paradigms, including participant surveys [5, 17, 36–38] and open-ended model exploration [39], can reveal salient failures in concept rendering, yet they do not constitute a scalable and repeatable methodology for comparing blindspot structure across model families.

This work. Motivated by these limitations, we argue that the identification and analysis of conceptual blindspots requires an automated, unsupervised methodology capable of eliciting concepts in the data distribution whose probability of occurrence under the true data-generating process diverges from their probability of occurrence under a trained generative model. Our contributions are three-fold:

- **Formalizing Conceptual Blindspots in Generative Image Models.** We introduce a systematic framework for identifying and quantifying conceptual blindspots in generative image models relative to natural images (Section 3). This formalization moves beyond anecdotal evidence and human-specified concept lists, thereby providing a principled basis for analyzing model limitations.
- **A Scalable, Unsupervised Approach for Identifying Conceptual Blindspots using Sparse Autoencoders.** We develop a structured methodology that uses sparse autoencoders (SAEs) to extract and compare concept distributions across natural and synthesized images (Section 4). To instantiate this methodology, we train and open-source an archetypal SAE (RA-SAE) on DINOv2 features with 32, 000 concepts, which is the largest RA-SAE of this kind to date.
- **Exploratory Tool and Analysis.** We provide an exploratory web tool that supports both distribution- and datapoint-level analyses of blindspots across models (Sections 5.1–5.4). Applying the method to Kandinsky, PixArt, and Stable Diffusion (SD) 1.5/2.1 (Section 5), we identify concrete instances of suppressed conceptual blindspots (e.g., bird feeders, DVD discs, and whitespaces on documents) as well as exaggerated conceptual blindspots (e.g., wood background texture and palm trees), as illustrated in Fig. 1.

The model, code, and web tool are available through a software repository.

¹We borrow the term “blindspots” from psychology literature [27], wherein it is used to refer to scenarios where an agent makes biased decisions despite exposure to observations that contradicts their decision’s rationale.

74 2 Related Work

75 2.1 Explainability in Vision

76 Early work in explainable AI for computer vision concentrated primarily on attributing predictions
77 to influential input regions [40, 42]; however, because such methods often supplied only limited
78 semantic information about learned representations and could yield incorrect explanations [43–
79 45], concept-based interpretability [46] emerged as a complementary paradigm for identifying
80 semantically meaningful directions in neural networks and thereby revealing not only where models
81 attend, but also which concepts and structures they operationalize [47–49].

82 Recent work further shows that several prominent concept-based interpretability methods—ACE [44],
83 CRAFT [48], and Sparse Autoencoders (SAEs) [50, 51]—can be understood as variants of a common
84 dictionary-learning problem instantiated under different constraints [52]. Within this family, sparse
85 autoencoders (SAEs) have become especially attractive for scalable concept-based interpretability,
86 although subsequent analyses have exposed limitations of original SAE formulations, including
87 excessively specific features [53], difficulties with compositional structure [54], and limited causal
88 intervention effects [55]; these limitations have motivated refined variants, among them the archetypal
89 SAE (RA-SAE) [56].

90 Beyond SAEs, interpretability approaches such as prompt-based probing [57], attention-map or acti-
91 vation visualization [58, 59], and dataset-level statistics [32, 33], including diversity and distribution-
92 coverage metrics, provide only partial purchase on the questions we study. Their central limitation
93 is one of granularity: they typically operate over entire images or prompts rather than over fine-
94 grained features and concepts [34, 35]. Moreover, because these approaches frequently depend on
95 subjective interpretation and do not discriminate among distinct failure modes [60], they remain
96 insufficient for the systematic identification of feature- and concept-level weaknesses in generative
97 image models [28].

98 2.2 Generative Image Models

99 Diffusion-based methods have become the dominant paradigm across multiple modalities in genera-
100 tive vision modeling, including image [4, 61–64], video [65–72], and 3D generation [7, 73–75]. In
101 image synthesis specifically, this trajectory can be traced to denoising diffusion probabilistic models
102 (DDPMs) [76], which were subsequently generalized to non-Markov diffusion processes through
103 denoising diffusion implicit models (DDIMs) [77].

104 The Stable Diffusion (SD) [9] family made DDPMs broadly accessible to both research and open-
105 source communities, after which the original SD model was followed by successive versions, includ-
106 ing SD 2 [78], SD 3 [79], and SD XL [80]. Numerous architectural modifications and extensions have
107 since introduced additional constraints on the diffusion process, including style [81, 82], pose [83],
108 and identity [84, 85], while also accommodating alternative input modalities such as image-to-image
109 generation; in parallel, Latent Diffusion Models built on or around the SD ecosystem, including
110 Kandinsky [86], PixArt [87], DeepFloyd [4, 88], and FLUX [89], have further diversified the model
111 landscape.

112 2.3 Datasets for Generative Image Models

113 The recent success of generative vision models is commonly attributed to the conjunction of expanded
114 computational resources and massive internet-scale datasets [90–92]. Among these resources, LAION-
115 5B [93] has been especially consequential for open-source text-to-image systems, including SD and
116 its derivatives; scraped from Common Crawl [94], it contains more than 5B image-caption pairs, 2.3B
117 of which are in English, while other influential corpora include COYO-700M [95] and Conceptual
118 Captions [96], with 700M and 3M image-caption pairs, respectively.

119 As LAION-5B became widely used, however, concerns regarding its biases and harmful content
120 intensified [97–101]. Despite filtering efforts, problematic material remained [97–99], including
121 NSFW content [100] and hundreds of CSAM instances [101], which ultimately prompted its tem-
122 porary removal from official channels; additionally, the dataset contains low-quality imagery [102]
123 and internet-style captions, such as product descriptions, whose linguistic form can diverge from the
124 prompts users supply to trained models [103].

2.4 Concept Discovery and Sparse Coding in Generative Image Models

Dictionary learning seeks sparse representations of input data in which each sample can be reconstructed as a linear combination of relatively few dictionary atoms [104–106]. Rooted in compressed sensing theory [107–108], the field has progressed from early vector-quantization methods [109] toward more elaborate formulations, including Non-negative Matrix Factorization [110–111], Sparse PCA [112], and K-SVD [113]; more recent developments include online methods [114], structured sparsity [115], theoretical recovery guarantees [116–117], and increasingly explicit connections to deep learning [118–119].

Advances in sparse coding have likewise been used to study the emergence of high-level concepts within diffusion models [120]. Before the diffusion era, concept-grounded interpretability was applied to earlier generative architectures through concept-bottleneck models, which require human intervention during training [121], and through post-hoc detectors that retrofit concept supervision [122]; however, because both strategies depend on human-defined concept inventories, they are intrinsically unable to expose broader trends that users have not explicitly anticipated.

Aggregate metrics, including precision, recall, density, and coverage [35, 123], together with latent-density scores that estimate sample quality from the model’s latent space [124], have become important tools for evaluating generative image models. Although such measures can reveal distributional gaps, they provide comparatively little information about which particular concepts are under- or over-represented.

Taken together, these limitations motivate a scalable and unsupervised framework capable of systematically identifying and quantifying concept-level failure modes in generative image models [125].

3 Formalizing Conceptual Blindspots in Generative Models

We begin by formalizing *conceptual blindspots* as systematic discrepancies between the conceptual content of natural images and that of model-generated outputs. This formalization permits the derivation of principled quantitative measures for determining which concepts are under- or over-represented by a generative model relative to its data distribution, as illustrated in Fig. 2. Although the construction relies on standard assumptions from prior work [126–130], our empirical results indicate that meaningful phenomenology continues to emerge even when these assumptions are only approximately satisfied.

Definition 1 (Data-Generating Process). Let $\mathcal{C} \subset \mathbb{R}^K$ denote a latent space with a Boltzmann prior $p(c) = \exp(-E(c))Z^{-1}$, where $E(\cdot)$ denotes an energy function that linearly decomposes over individual latents and Z is the corresponding partition function, i.e., $E(c) = \sum_k E(c_k)$ and hence $p(c) = \prod_k p_k(c_k)$. A data-generating process (DGP) is an invertible function $G: \mathcal{C} \rightarrow \mathcal{X}$ that maps the latents $c \in \mathcal{C}$ to observations $x \in \mathcal{X}$, i.e., $x = G(c)$.

For notational economy, we use $p(\cdot)$ to denote both the latent density $p(c)$ and its pushforward to image space $p(x)$, where $x = G(c)$; this convention is valid because the assumed invertibility of G induces a well-defined distribution over $\mathcal{X}$ through the change-of-variables formula. In this formulation, the individual dimensions of the latent space correspond to the **Concepts** underlying the data distribution $P_{\mathcal{X}}$ over the observation space of images $\mathcal{X}$. Different latent coordinates may, for instance, encode concepts such as color, shape, size, location, and related factors of variation [131–132]. We additionally allow the data-generating process to associate a text description $t \in \mathcal{T}$ with each image sampled from the data distribution, although we do not explicitly model this mechanism. These descriptions can then be used to train a text-conditioned **Generative image model** g_θ , with parameters θ , on image-text pairs so that it maps a noise signal $\eta \sim \mathcal{N}(\mathbf{0}, I)$ and a scene description t to an image x depicting that description.

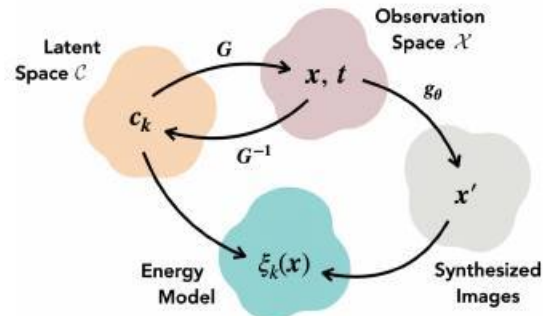


Figure 2: **Intuitive mapping of our framework.** Latent concepts $c_k \in \mathcal{C}$ are mapped to observations (x, t) by the invertible DGP, while S_θ generates images x' conditioned on t . The energy model $\xi_k(x)$ extracts concept representations from both x and x' , thereby enabling distributional comparison and blindspot identification.

To define conceptual blindspots in g_θ , we compare the probability mass assigned to a concept by the data-generating process with the corresponding mass induced by the model. Accordingly, we begin with an evaluation set of natural image-text pairs $(\mathcal{D}_\mathcal{X}, \mathcal{D}_\mathcal{T})$ and construct a companion set of synthetic images $\mathcal{D}'_\mathcal{X}$ from the associated text descriptions. To estimate concept occurrence probabilities, we introduce an **Energy model** $\xi : \mathcal{X} \rightarrow \mathbb{R}^d$ that maps each image to a d -dimensional vector whose k^{th} coordinate is a scalar estimate of the energy associated with the k^{th} concept, i.e., $\xi_k(\mathbf{x}) = E(c_k)$. These estimates are meaningful under the assumed invertibility of the data-generating process. The probability that the data-generating process assigns to an image $\mathbf{x}$ can therefore be written as $p_k(\mathbf{x}) \propto \exp(-\sum_k \xi_k(\mathbf{x}))$ (where Z_ξ is the partition function), yielding the population-level estimate $p_k(\mathcal{D}_\mathcal{X}) = \prod_{\mathbf{x} \in \mathcal{D}_\mathcal{X}} p_k(\mathbf{x})$. Using this estimate together with the sigmoid map $\sigma(\cdot)$, we define the average energy difference for the k^{th} concept between $\mathcal{D}_\mathcal{X}$ and $\mathcal{D}'_\mathcal{X}$.

Definition 2 (Energy Difference). Let $\mathbf{x} \in \mathcal{D}_\mathcal{X}$ denote a real image sampled from the data-generating process G , and let $\mathbf{x}' \in \mathcal{D}'_\mathcal{X}$ be a synthetic image generated by the model g_θ . Let $\xi_k : \mathcal{X} \rightarrow \mathbb{R}$ denote the energy assigned to the k^{th} concept by the energy model ξ . We define the energy difference for concept k as:

$$\begin{aligned} \delta_{g_\theta \leftrightarrow G}(k) &= \sigma(\mathbb{E}_{\mathbf{x}'}[\xi_k(\mathbf{x}')] - \mathbb{E}_{\mathbf{x}}[\xi_k(\mathbf{x})]) \\ &= \frac{p_k(\mathcal{D}'_\mathcal{X})}{p_k(\mathcal{D}_\mathcal{X}) + p_k(\mathcal{D}'_\mathcal{X})}, \end{aligned} \quad (1)$$

where the expectations are taken over $\mathcal{D}'_\mathcal{X}$ and $\mathcal{D}_\mathcal{X}$, respectively, and $p_k(\mathcal{D}) \propto \exp(-\sum_{\mathbf{x} \in \mathcal{D}} \xi_k(\mathbf{x}))$ denotes the unnormalized conceptual probability mass of dataset $\mathcal{D}$ under concept k .

Thus, the energy difference for the k^{th} concept expresses the relative probability with which a concept occurs in an observation set, here $\mathcal{D}'_\mathcal{X}$, compared with a baseline dataset, here $\mathcal{D}_\mathcal{X}$. This quantity directly supports the following definition of conceptual blindspots.

Definition 3 (Suppressed / Exaggerated Conceptual Blindspots). Given a generative image model g_θ , we say, compared to the data-generating process G , c_k is a suppressed conceptual blindspot in the model if $\delta_{g_\theta \leftrightarrow G}(k) < \lambda_{\min}$ and exaggerated if $\delta_{g_\theta \leftrightarrow G}(k) > \lambda_{\max}$.

Overall, a conceptual blindspot is a concept whose likelihood of appearing in generated images departs substantially, through either suppression or exaggeration, from its prevalence under the data-generating process. Suppressed concepts exhibit disproportionately low activation (e.g., $\delta(k) < \lambda_{\min}$), whereas exaggerated concepts are overrepresented (e.g., $\delta(k) > \lambda_{\max}$); throughout the empirical analysis, we set $\lambda_{\min} = 0.1$ and $\lambda_{\max} = 0.9$ to isolate these regimes.

This definition is related to the notion of “mode collapse” studied in prior work (e.g., see Bau et al. [39]), but it differs in the granularity at which model failure is measured. Whereas mode collapse concerns exaggerated or suppressed odds of generating *entire images*, our analysis concerns altered odds of generating particular concepts. For example, if a model fails to produce images of an object with a white background, we describe that concept as a suppressed conceptual blindspot.

4 Method: Operationalizing the Definition of Conceptual Blindspots

We now describe the operational pipeline through which conceptual blindspots are identified for a generative model g_θ . Following Sec. 3 the required ingredients are (i) a collection of samples from g_θ that can be paired with observations from the ground-truth generative process and (ii) an energy model capable of mediating the comparison between the two distributions. Below, $\|\cdot\|_F$ denotes the Frobenius norm and $\|\cdot\|_0$ denotes the number of non-zero entries (the ℓ_0 pseudo-norm). For a vector or matrix $\mathbf{X}$, $\mathbf{X} \geq 0$ indicates element-wise non-negativity. For $n > 0$, we write $[n] := \{1, \dots, n\}$ and denote the i -th row of a matrix $\mathbf{A}$ by $\mathbf{A}_i$.

From Prompts to Latent Representations. To identify conceptual blindspots in a model g_θ , we compare a dataset $\mathcal{D}_\mathcal{X}$ of image-caption pairs $(\mathbf{x}, \mathbf{t})$ sampled from the data-generating process G with synthetic counterparts produced by the generative model g_θ using the same text descriptions.

Specifically, given $\mathbf{t}$, we synthesize a counterpart image $\mathbf{x}'$ using a pretrained text-to-image generator $g_\theta : \mathcal{T} \rightarrow \mathcal{X}$, instantiated as a denoising diffusion probabilistic model (DDPM) [78, 86, 87].

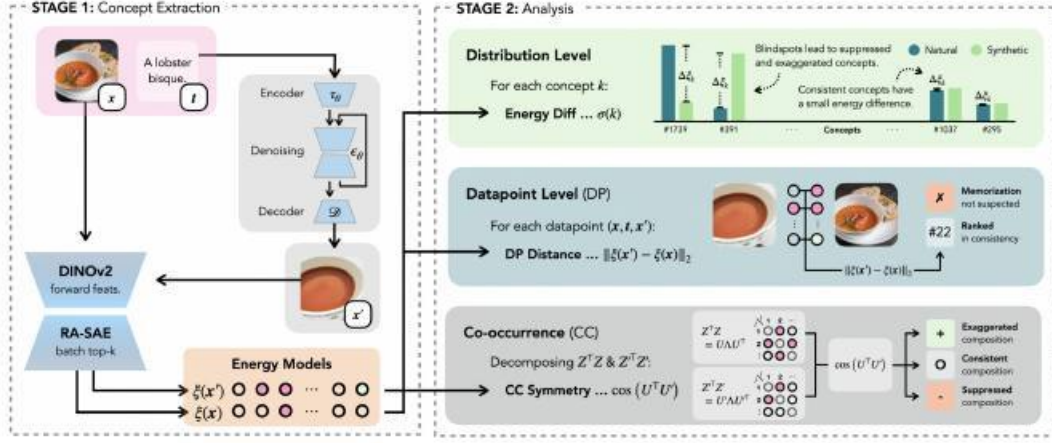


Figure 3: **Concept Extraction Pipeline.** For a triplet (x, t, x') , DINOv2 features are extracted from each image and then processed by a RA-SAE into sparse concept embeddings, yielding energy models $\xi(x)$ and $\xi(x')$. In particular, $\xi_k(x) = E(c_k)$ represents the energy assigned to the k^{th} concept.

226 Sampling proceeds in latent space through the reverse trajectory $(\gamma_t)_{t=0}^T$:

$$\gamma_T \sim \mathcal{N}(\mathbf{0}, \mathbf{I}), \quad \gamma_{t-1} = \frac{1}{\sqrt{\alpha_t}} \left(\gamma_t - \frac{1-\alpha_t}{\sqrt{1-\alpha_t}} \epsilon_\theta(\gamma_t, t, c) \right) + \sigma_t \eta_t, \quad \text{and} \quad \eta_t \sim \mathcal{N}(\mathbf{0}, \mathbf{I}),$$

227 where $\alpha_t \in (0, 1)$ and $\bar{\alpha}_t = \prod_{s=1}^t \alpha_s$ follow the standard cosine noise schedule. The final latent γ_0
 228 is decoded via a pretrained VAE to produce the synthetic image $x' = \text{VAE}(\gamma_0)$. In the remainder of
 229 the paper, we therefore treat g_θ as a black box implementing the prompt-to-image map $t \mapsto x'$.

230 **Defining the Energy Model.** Building on prior work showing that self-supervised learning methods
 231 can invert the data-generating process and identify its underlying energy function up to linear
 232 transformations [126, 128, 133, 134], we use DINOv2 [135], a state-of-the-art self-supervised model,
 233 as the basis for our analysis. Because the number of concepts underlying the DGP is expected to
 234 exceed the dimensionality of the model’s feature space [51, 136], we train sparse autoencoders (SAEs)
 235 on DINOv2 features in order to identify subspaces corresponding to those concepts [50, 56, 137–139].
 236 The guiding intuition is that, if DINOv2 represents the concepts underlying the generative process
 237 through approximately orthogonal directions, as assumed by the independence constraint in Def. 1,
 238 then an SAE should be able to separate those concepts into individual latent dimensions [136].
 239 The activation assigned by the SAE to each dimension then serves as our approximation to the
 240 ground-truth energy function for the concept represented by that dimension.

241 Formally, with $f : \mathcal{X} \rightarrow \mathbb{R}^d$ denoting the DINOv2 feature extractor, we obtain features $a =$
 242 $f(x) \in \mathbb{R}^d$ from both natural and synthetic images in $\mathcal{D}_\mathcal{X}$ and $\mathcal{D}'_\mathcal{X}$. For n image-text pairs, the
 243 real and generated features are stacked into matrices $\mathbf{A}, \mathbf{A}' \in \mathbb{R}^{n \times d}$. Each feature vector is then
 244 decomposed into a sparse combination of learned concept atoms using an SAE. Let $\mathbf{D} \in \mathbb{R}^{d \times K'}$
 245 denote a dictionary of K' concept vectors, and let $\Psi : \mathbb{R}^d \rightarrow \mathbb{R}^{K'}$ be the SAE encoder mapping input
 246 features to sparse codes. Applying Ψ row-wise gives the activation matrix $\mathbf{Z} = \Psi(\mathbf{A}) \in \mathbb{R}^{n \times K'}$,
 247 where each row $z_i = \Psi(a_i)$ represents an image-level decomposition over learned concepts. The
 248 decoder reconstructs features as $\mathbf{Z}\mathbf{D}^\top$, and the SAE is trained to minimize reconstruction error
 249 subject to sparsity and non-negativity:

$$\min_{\Psi, \mathbf{D}} \|\mathbf{A} - \Psi(\mathbf{A})\mathbf{D}^\top\|_F^2 \quad \text{s.t.} \quad \Psi(\mathbf{A}) \geq 0, \|\Psi(\mathbf{A})_i\|_0 \ll K' \quad \forall i \in [n]. \quad (2)$$

250 Vanilla SAEs often converge to arbitrarily oriented dictionaries, making downstream analyses undesir-
 251 ably sensitive to random initialization. To mitigate this instability, and thereby improve reproducibility
 252 and seed independence, we employ the *Archetypal* SAE (RA-SAE) [56] with a TOP-K sparsity con-
 253 straint [137]. RA-SAE constrains the dictionary $\mathbf{D}$ to lie within the convex combinations of the
 254 training data. Specifically, we write $\mathbf{D} = \mathbf{W}\mathbf{A}$ with $\mathbf{W} \in \Omega_{K', n}$, the set of row-stochastic matrices
 255 in $\mathbb{R}^{K' \times n}$:

$$\Omega_{K', n} := \left\{ \mathbf{W} \in \mathbb{R}^{K' \times n} \mid \mathbf{W} \geq 0, \mathbf{W}\mathbf{1} = \mathbf{1} \right\}. \quad (3)$$

Consequently, every atom $\mathbf{D}_i$ lies in the convex hull of the data $\text{conv}(\mathbf{A})$, and every reconstruction $\mathbf{Z}\mathbf{D}^\top$ remains inside the conic hull of the data $\text{cone}(\mathbf{A})$. This geometric constraint helps ensure that learned concepts remain faithful to the support of the data distribution [56]. After training, the SAE provides a common and stable set of sparse codes, $\mathbf{Z}$ for real images and $\mathbf{Z}'$ for their generated counterparts. *These codes express the same prompt-conditioned visual semantics in a shared interpretable basis, with each concept activation functioning as an energy value for our analysis of conceptual blindspots.* Thus, given a prompt and its associated real image (t, x) , our method produces two sparse concept vectors (z, z') , enabling direct comparison between real and generated visual content in a shared conceptual basis.

In summary, the full pipeline begins with a captioned image (t, x) , synthesizes a counterpart x' , and maps both images through a vision encoder and trained SAE into a shared sparse concept space, yielding interpretable representations (z, z') that form the basis for measuring conceptual shifts induced by the generative process.

5 Results

We analyze four generative image models trained on LAION-5B—SD 1.5, SD 2.1, PixArt, and Kandinsky—using $|\mathcal{X}| = 10,000$ image-text pairs and their corresponding generations (Fig. 4). Our analysis operates at three complementary levels (Fig. 3): a **distribution-level** evaluation characterizes suppressed and exaggerated concepts; a **datapoint-level** analysis exposes failures associated with ambiguity, omission, and memorization; and a **compositional** analysis identifies subtler distortions in concept co-occurrence geometry.

A central contribution of this work is the interactive exploratory tool shown in Fig. 5. The tool is a web-based interface organized around a UMAP projection of concept representations, allowing users to visualize and compare concept-level energy differences across models; it is publicly available through a software repository, together with pre-computed energy-difference data for the four evaluated models (SD 1.5, SD 2.1, PixArt, and Kandinsky). The analyses reported below are derived from observations enabled by this tool, whose principal functionalities include:

- **Contrast different models and architectures.** For each evaluated model, the tool provides a UMAP visualization over all 32,000 RA-SAE concepts, with each scatter point representing an individual concept and its color encoding the corresponding energy difference.
- **Inspect concepts.** Each concept is accompanied by a detailed card containing key statistics, representative real and generated images (x, x') , and visualizations of co-occurrence patterns.
- **Explore blindspots.** Beyond UMAP and per-concept views, the interface includes global rankings of suppressed and exaggerated blindspots, thereby foregrounding the most salient conceptual discrepancies.

Rather than exhaustively analyzing a single phenomenon, we report a set of high-level findings that demonstrate the tool’s versatility and support broader, user-directed exploration.

5.1 Generative Image Models Cannot Produce Unseen Concepts

To assess disparities between the generative models and the natural image distribution, we first analyze the marginal energy difference $\delta(k)$ across the 32,000 concepts learned by the RA-SAE. As defined in Sec. 4, this quantity measures the relative prevalence of each concept in synthesized images compared with natural images: $\delta(k) < 0.1$ indicates that concept k is under-represented, or suppressed, in generated images, whereas $\delta(k) > 0.9$ indicates over-representation, or exaggeration. Fig. 6 reports the distribution of $\delta(k)$ for each of the four evaluated models. Across models, the resulting histograms are heavy-tailed and allocate substantial mass to both extremes, indicating systematic discrepancies in concept coverage rather than isolated outliers. The left tail, which corresponds to suppressed concepts,



Figure 4: **Representative image-caption pairs** from LAION-5B, together with corresponding images generated from the same prompt by SD 1.5, SD 2.1, PixArt, and Kandinsky.

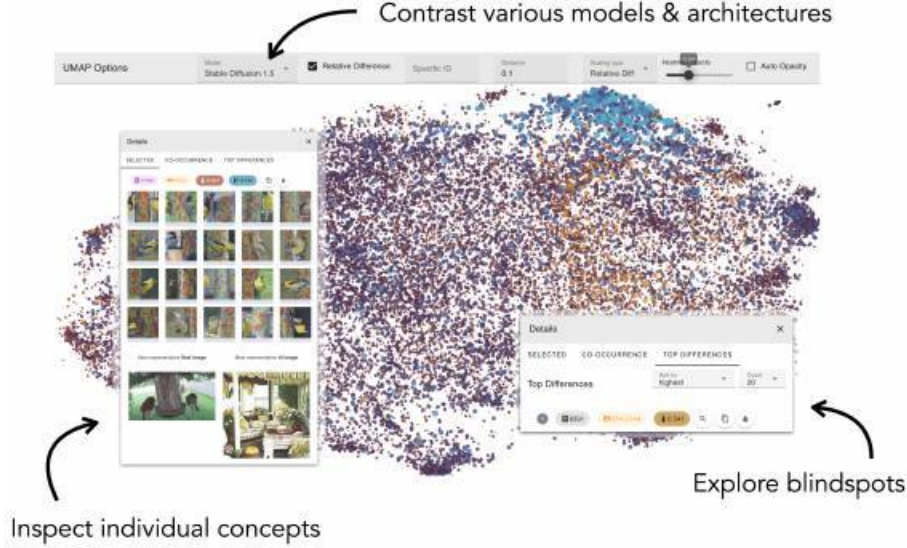


Figure 5: **Overview of the Exploratory Tool.** The web interface presents a UMAP projection for each evaluated model, with each dot denoting a concept and color indicating its energy difference. Upon concept selection, a detail panel displays illustrative images, summary statistics, and representative natural and generated images x and x' , while ordered co-occurrence lists and global blindspot rankings support further inspection.

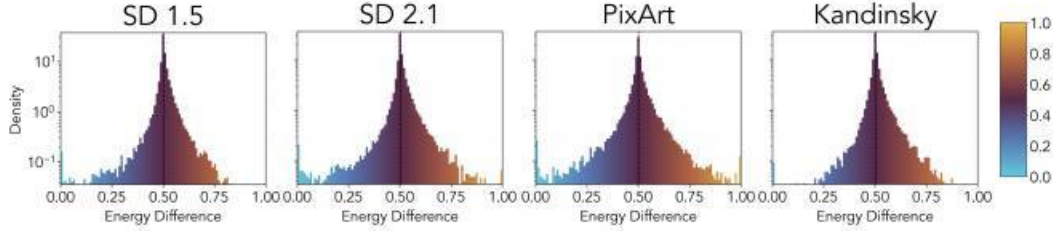


Figure 6: **Concept Energy Distribution.** Log-scale histograms of energy differences $\delta(k)$ over 32,000 concepts, comparing natural and synthesized distributions for each evaluated model. Values to the left of zero indicate suppressed, under-represented concepts, whereas values to the right indicate exaggerated, over-represented concepts.

is especially dense and extended relative to the right tail, revealing a persistent tendency toward concept suppression. This asymmetry is also reflected in the negative skewness of the distributions: Skewness = -0.54 for SD 2.1, -0.40 for both SD 1.5 and PixArt, and -0.23 for Kandinsky.

Although every model exhibits both suppressed and exaggerated concepts, their blindspot profiles are not identical. PixArt, for example, displays a wider spread, consistent with a more suppressed concept distribution. Nevertheless, the uniform left-skew across all four distributions suggests a shared propensity for concept omission, even though the detailed structure of this behavior requires the more specific analyses presented in the following sections.

5.2 Structure and Specificity of Conceptual Blindspots

Having quantified marginal discrepancies in concept frequency, we next examine their global organization by embedding the full set of 32,000 sparse-code concepts into two dimensions with UMAP and coloring each concept according to its $\delta(\cdot)$ value. As Fig. 7 shows, visually coherent clusters emerge for all evaluated models. These clusters frequently align with contiguous regions of conceptual blindspots, particularly among suppressed (blue) concepts, suggesting that blindspots are not randomly dispersed but instead reflect structured biases arising from training distributions, architectural priors, or their interaction. To evaluate the cross-model consistency of these patterns, we further analyze both the magnitude and structure of concept-level $\delta(\cdot)$ values. Fig. 8 presents

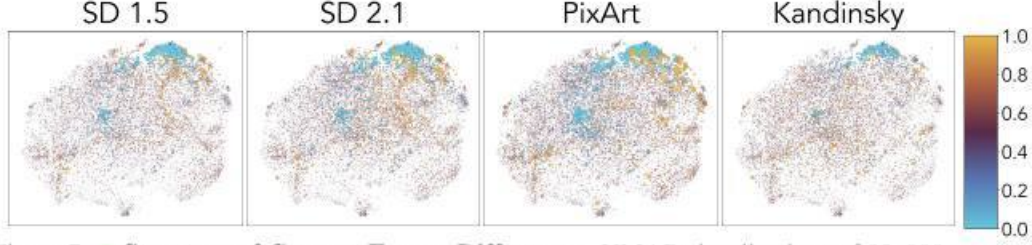


Figure 7: **Structure of Concept Energy Differences.** UMAP visualizations of 32,000 concepts, colored by their energy difference $\delta(k)$ between natural and synthesized distributions. The resulting clusters reveal structured blindspot patterns, with suppressed concepts concentrated toward the blue end and exaggerated concepts toward the yellow end.

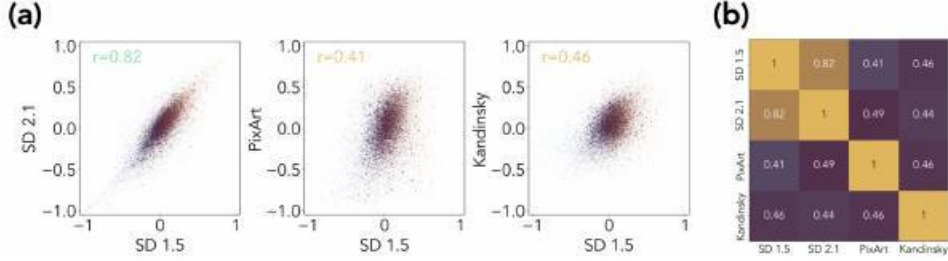


Figure 8: **Cross-Model Concept Energy Correlation.** Pairwise scatter plots of $\delta(k)$ across all four evaluated models, with Pearson correlation coefficients reported in the upper-left corner. Strong alignment between SD 1.5 and SD 2.1 contrasts with weaker correlations involving other architectures, indicating model-specific blindspot structure. **Correlation Matrix of Conceptual Blindspots.** The heatmap summarizes pairwise Pearson correlations for δ across all models, quantifying the extent to which conceptual blindspots are shared.

scatter plots and pairwise Pearson correlation coefficients comparing the $\delta(k)$ vectors of SD 1.5 against those of the other models. As expected, SD 1.5 and 2.1 are strongly correlated ($r = 0.82$), consistent with their shared architectural and training pipelines. By contrast, correlations with PixArt and Kandinsky are markedly lower— $r = 0.41$ and $r = 0.46$, respectively—indicating that distinct architectures emphasize different regions of conceptual space.

Overall, this analysis indicates that some blindspots are broadly shared, plausibly reflecting properties of the underlying dataset, whereas others are highly model-specific and may arise from idiosyncratic training dynamics or differences in model capacity. This observation motivates the subsequent distinction between blindspots that recur across models and those that appear unique to particular model families.

5.3 Qualitative Examples of Blindspots

We next visualize concrete examples of suppressed and exaggerated blindspots in order to clarify which concepts occupy these regimes. In Fig. 9a, we show a conceptual blindspot suppressed by all evaluated models: none successfully reproduces the concept `solid white on documents`. Although the caption explicitly invokes this concept, the generated images fail to realize the intended visual semantics, suggesting that this region of concept space is systematically under-sampled across models. Fig. 9b, by contrast, illustrates a model-specific blindspot: the concept `pan` is captured by three models but is conspicuously absent from Kandinsky generations. This observation reinforces the results in Sec. 5.2 where cross-model agreement was substantial for some blindspots but markedly weaker for others.

Conversely, Fig. 10 presents an exaggerated blindspot in which the concept `shadow under animal` is overemphasized in generated images. Although shadows are locally plausible, their consistently pronounced rendering across models, especially relative to their more nuanced and variable appearance in natural images, suggests the presence of an overactive generative prior. Notably, despite searching for concepts uniquely exaggerated by individual models, we did not identify clear examples, which suggests that the most salient exaggerations in this setting are approximately universal.

Taken together, these examples demonstrate how conceptual blindspots materialize in generated outputs and show that the energy-based diagnostic can expose both shared and model-specific failure modes. In particular, the method surfaces unexpected limitations, such as the consistent failure across all models to reproduce clear or solid background elements like whitespaces on documents, thereby raising the possibility that some failure patterns arise from architectural constraints or training-data biases that transcend individual model idiosyncrasies.

Although these aggregate analyses are informative, they also motivate a finer-grained question: do blindspots appear only after aggregating across many samples, or are they already visible at the level of individual datapoints? This perspective allows us to probe the mechanisms underlying blindspots more directly, including prompt misinterpretation, latent memorization, or both.

5.4 Investigating Datapoint-level Energy Difference: From Incongruent to Memorized Images

To move beyond population-level statistics, we examine individual natural-versus-generated image pairs for which the $\delta(\cdot)$ values, averaged over all concepts, exhibit the smallest and largest discrepancies. This analysis makes model successes and failures more interpretable, with the latter often arising from prompt ambiguity or memorization artifacts. For example, Fig. 11a shows instances whose average $\delta(\cdot)$ values are nearly zero. In these cases, the generated images are conceptually difficult to distinguish from the originals. However, qualitative inspection indicates that this apparent alignment is not necessarily due to faithful prompt semantics, but can instead reflect the replication of memorized templates: repet-



Figure 10: **Example of an Exaggerated Conceptual Blindspot.** Four synthesized images $D'_{\mathcal{X}}$, archetypal for the concept shadow under animal, are displayed alongside the corresponding natural image $D_{\mathcal{X}}$ and generation caption t . The accentuated, nearly solid black shadows beneath the animals make this blindspot universal across the evaluated models.

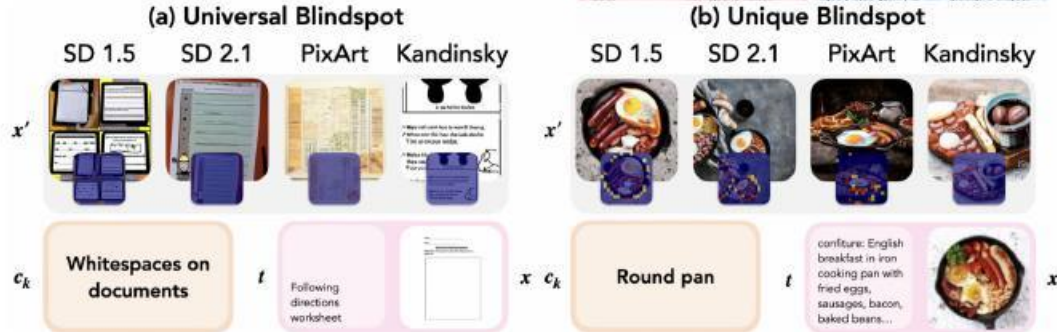


Figure 9: **Examples of Suppressed Conceptual Blindspots.** Natural images x , representative concepts c_k , and captions t are shown alongside four synthesized images x' generated with S_θ . The universal blindspot appears in all evaluated models, whereas the unique blindspot appears only in Kandinsky.

391 itive visual structures, such as clothing outlines
 392 or object arrangements, suggest copying from
 393 overly frequent training patterns. By contrast,
 394 Fig. 11b shows samples with among the largest
 395 $\delta(\cdot)$ values. These examples exhibit substan-
 396 tial conceptual divergence between synthesized
 397 and natural images. Although some discrepan-
 398 cies are attributable to underspecified or noisy
 399 captions, others reveal genuine blindspots: the
 400 prompt specifies a clear concept that is faithfully
 401 present in $\mathcal{D}_{\mathcal{X}}$, yet the model fails to realize it in $\mathcal{D}'_{\mathcal{X}}$. Such failures suggest that, even when language
 402 grounding is adequate, certain concepts remain outside the model’s generative capabilities.

403 5.5 Analyzing Post-Training Effects

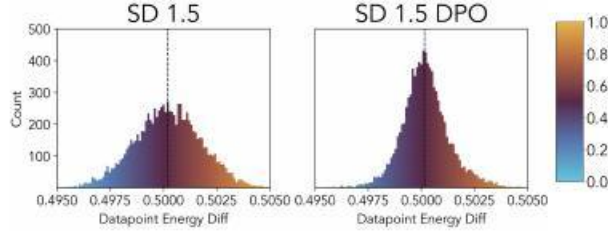


Figure 12: **Effect of DPO on Concept Fidelity.** Histogram of datapoint-wise energy differences between synthesized and natural distributions for SD 1.5 models trained with and without DPO.

404 Post-training protocols, including safety fine-tuning, have been argued to reduce the diversity of
 405 model generations [140]. Because our pipeline can isolate salient discrepancies between a model’s
 406 generations and the ground-truth DGP, we use it here to examine the effects of DPO, a widely used
 407 safety fine-tuning protocol [141]. Specifically, we compare two variants of SD 1.5, one trained
 408 with DPO and one trained without it. For each image pair $(\mathcal{D}_X, \mathcal{D}'_X)$, we compute the ℓ_2 norm of
 409 the difference between their internal concept-energy vectors, $\|\xi(\mathcal{D}'_X) - \xi(\mathcal{D}_X)\|_2$. Fig. 12 shows
 410 a histogram of these datapoint-wise energy differences. The DPO-enhanced model has both a
 411 lower median and a narrower spread, indicating that its generated concepts are more consistently
 412 distributed with respect to the ground-truth DGP. This pattern suggests that DPO may regularize the
 413 model’s concept distribution by encouraging outputs whose semantic content more closely reflects the
 414 observed inputs. Although our analysis does not disentangle the particular inductive biases introduced
 415 by DPO, the results provide empirical evidence that its human-preference-oriented optimization
 416 objective indirectly improves agreement with the training distribution. In particular, it reduces *both*
 417 *over and under* activation of individual concepts relative to the natural baseline. These findings
 418 illustrate the utility of our pipeline for characterizing post-training interventions not only through
 419 output quality, but also through their effects on the conceptual geometry of the model’s output space.

420 5.6 Conceptual Misalignment as a Function of Empirical Frequency

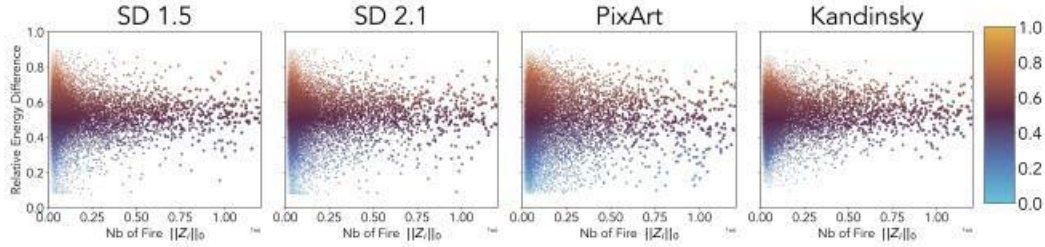


Figure 13: **Concept Fidelity Across Frequency Spectrum.** Scatter plots depict the relationship between concept frequency (x-axis) and energy difference (y-axis) for the four evaluated models. Each point denotes a concept, with marker size proportional to its activation frequency.

421 We hypothesized above that conceptual blindspots are not merely architectural artifacts, but may
 422 also arise from distributional peculiarities associated with particular features. Here, we empirically
 423 test this hypothesis by asking whether concepts that are rarely activated in natural images, namely
 424 those occupying the long tail of the data distribution, are especially likely to become blindspots
 425 in generative models. Concretely, we process the natural dataset $\mathcal{D}_X$ through the trained SAE and
 426 compute, for each concept k , its empirical frequency $\|Z_{:,k}\|_0$, where $Z_{:,k}$ denotes the activations of
 427 concept k across all images. We then correlate this frequency with the absolute energy difference
 428 observed in generated outputs. Fig. 13 visualizes the resulting relationship for all evaluated models.
 429 Concepts that occur more frequently in natural data tend to exhibit smaller energy discrepancies,
 430 whereas rare concepts, particularly suppressed ones ($\delta(k) < 0.5$), show substantial alignment errors.
 431 This pattern suggests that many blindspots arise not from random model quirks, but from systematic
 432 effects associated with long-tail concept distributions, and that mitigating them may require data
 433 reweighting or augmentation in addition to architectural changes.

6 Discussion

Our analysis reveals multiple conceptual blindspots across four widely used generative image models. The findings reported here, however, should be understood as an initial characterization rather than an exhaustive account: each observed phenomenon could support a dedicated investigation in its own right. Rather than pursuing any single thread to completion, we have emphasized the breadth and flexibility of the proposed method and exploratory tool, which together enable systematic identification of difficult-to-generate concepts, detection of memorization artifacts, discovery of datapoints with insufficient captions, quantification of post-training effects, and characterization of conceptual shifts across model architectures. This framing opens several directions for follow-up work, including deeper analyses of particular blindspots, broader evaluation across additional architectures, and more detailed investigation of the causal mechanisms underlying conceptual failures. Future work could also develop hierarchical RA-SAE concept representations, which would permit more nuanced analyses of concept organization and compositional failure.

Limitations. Several limitations qualify the present analysis. Because concept extraction and representation rely on DINOv2 and RA-SAE, the framework is necessarily constrained by the concepts these models make available; concepts that are poorly represented in this feature-and-dictionary pipeline will remain outside the scope of detection. In addition, although the evaluation set of 10,000 images is substantial, it may still be insufficient to fully characterize rare long-tail concepts, higher-order concept co-occurrences, or more delicate compositional statistics.

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Appendix

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A Experimental Setup

This section specifies the experimental configuration used to analyze four widely adopted generative image models: SD 1.5/2.1, Kandinsky, and PixArt, each trained on LAION-5B or on subsets and derivatives of that corpus. The accompanying code is available through a software repository.

A.1 Observation Space

The observation space is constructed by sampling 10,000 image-text pairs from LAION-5B [93], which serves as the natural-image domain for our comparisons. Because concerns about CSAM and other unsafe content have made the original release unavailable, we instead use a substitute subset with additional unsafe-content filtering, available at <https://huggingface.co/datasets/laion/relaion2B-en-research-safe>

The sampling procedure comprises three steps: (1) loading the LAION dataset with the Hugging Face datasets library, (2) validating URL structure and resource availability through HTTP HEAD requests, and (3) repeatedly sampling with replacement until 10,000 valid examples are obtained. This procedure yields D_G as a collection of (x, t) tuples and ensures that the observation space contains accessible image-text pairs, which is important because a substantial fraction of the original image URLs has become unavailable since release. Additional synthesized examples appear in Appendix H

A.2 Synthesized Images

For each evaluated model, we construct a synthetic dataset D_{g_θ} in one-to-one correspondence with D_G , thereby producing triplets (x, x', t) . Concretely, for every one of the 10,000 image-text pairs (x, t) in D_G , the caption t is used as the prompt from which each generative model g_θ synthesizes a counterpart image x' .

The synthesis procedure follows the standard text-to-image pipeline for each architecture and is implemented with the Hugging Face diffusers library, with models loaded in mixed precision (fp16). Unless otherwise stated, all synthetic images are generated at 512×512 resolution using default parameters.

Stable Diffusion 1.5. We use the checkpoint at <https://huggingface.co/benjamin-paine/stable-diffusion-v1-5>, which mirrors the deprecated checkpoint at <https://huggingface.co/CompVis/stable-diffusion-v1-5>. Inference uses 50 steps with guidance scale fixed at 7.5.

Stable Diffusion 1.5 + DPO. The DPO variant of SD 1.5, used in Sec. 5.3 follows the baseline SD 1.5 implementation while replacing the UNet component with the DPO-trained version from <https://huggingface.co/mhdang/dpo-sd1.5-text2image-v1>

Stable Diffusion 2.1. We use the checkpoint from <https://huggingface.co/stabilityai/stable-diffusion-2-1>, inference is performed for 50 steps with guidance scale fixed at 7.5.

Kandinsky. We use the checkpoint from <https://huggingface.co/kandinsky-community/kandinsky-2-1>, inference is performed for 100 steps with guidance scale fixed at 4.0.

PixArt. We use the checkpoint from <https://huggingface.co/PixArt-alpha/PixArt-XL-2-1024-MS>, inference is performed for 50 steps with guidance scale fixed at 7.5.

A.3 Distribution Level Analysis

Section 5.1 For each evaluated model, we compute energy differences $\delta(\cdot)$ over all 32,000 concepts. During normalization, we apply a sigmoid transformation with temperature $T = 0.8$, and the resulting values are visualized as log-scale density histograms using 100 bins over $[0, 1]$.

Section 5.2 We embed the complete set of 32,000 sparse-code concepts into two dimensions with UMAP. Each UMAP point denotes an individual concept and is colored by its energy difference $\delta(\cdot)$,

927 making both suppressed and exaggerated blindspots visible. To quantify cross-model consistency, we
 928 compute pairwise Pearson correlations between the $\delta(\cdot)$ vectors of all model pairs, producing scatter
 929 plots and correlation matrices that reveal whether blindspots cluster in conceptual space and whether
 930 those clusters are model-specific or universal.

931 **Section 5.3** We rank all 32,000 concepts by their energy difference $\delta(\cdot)$ and manually inspect
 932 the extrema corresponding to suppressed and exaggerated blindspots. Suppressed blindspots are
 933 selected from concepts with $\delta(\cdot) < 0.1$, while exaggerated blindspots are selected from concepts
 934 with $\delta(\cdot) < 0.9$. The reported examples are manually annotated with textual descriptions obtained by
 935 inspecting the most activating images and associated spatial attention patterns. Ongoing efforts to
 936 automate this interpretation process are described in Appendix D

937 **Section 5.5** We compare SD 1.5 with and without DPO by computing, for each image pair (x, x') ,
 938 the L2 norm of the difference between their concept-energy vectors $\|\xi(x') - \xi(x)\|_2$. Before
 939 averaging, we apply a sigmoid transformation with temperature $T = 0.8$ to the element-wise
 940 differences, yielding datapoint-wise energy differences that quantify the conceptual deviation of each
 941 generated image from its natural counterpart. These differences are then visualized as overlapping
 942 histograms to contrast the two model variants.

943 **Section 5.6** For each concept c_k , we compute its empirical frequency $\|Z_{:,i}\|_0$, defined as the
 944 number of non-zero activations across the natural dataset. We then apply sigmoid normalization
 945 with temperature $T = 0.4$ to the energy differences $\delta(\cdot)$. The analysis is visualized with scatter
 946 plots in which the x-axis gives empirical concept frequency, the y-axis gives sigmoid-transformed
 947 energy difference, marker size is proportional to activation frequency, and marker color encodes the
 948 magnitude of the energy difference.

949 A.4 • Datapoint Level Analysis

950 **Section 5.4** For each image pair (x, x') , we compute the L2 norm of the difference between
 951 the corresponding concept-energy vectors, $\|\xi(x') - \xi(x)\|_2$, thereby obtaining a scalar measure of
 952 conceptual divergence. Samples are ranked by this energy difference: minimal divergence indicates
 953 potential memorization artifacts, whereas maximal divergence identifies substantial conceptual
 954 failures. This ranking enables targeted qualitative inspection of specific failure modes.

955 A.5 • Co-occurrence Analysis

956 **Appendix F** For the natural and synthesized datasets D_G and $D_{g\theta}$, we analyze concept co-
 957 occurrence through the co-activation matrix $Z^T Z$, which records pairwise correlations in concept
 958 usage. Spectral analysis via eigendecomposition is used to inspect dominant conceptual directions,
 959 and alignment between natural and synthetic co-occurrence structures is measured with cosine-
 960 similarity heatmaps between the top-100 eigenvectors of each co-occurrence matrix. These 100×100
 961 matrices are visualized as square heatmaps, where perfect diagonal structure would indicate identical
 962 principal concept axes and off-diagonal structure reveals rotations or mismatches in compositional
 963 geometry.

B Custom RA-SAE Training Details

This section summarizes the training configuration for our custom archetypal SAE (RA-SAE) [56]. The resulting model is released through a software repository.

Dataset. The auto-encoder is trained on the full ImageNet-1k training split, comprising approximately ≈ 1.28 M RGB images. Each image is converted into 261 visual tokens with DINOv2 [135], and these tokens are supplied to the SAE without class or positional embeddings. The total number of training tokens is therefore $50 \times 1.28\text{M} \times 261 \approx 1.67 \times 10^{10}$.

Dictionary. The dictionary contains 32,000 concept dimensions. Sparsity is imposed through top- k masking with $k = 5$, so activations outside the largest five for each input are set to 0. We initialize weights with Xavier/Glorot initialization, train primarily in mixed precision (fp16), and run the final ten epochs at full precision.

Optimizer and Schedule. The model is trained for 50 epochs with AdamW ($\beta_1 = 0.9$, $\beta_2 = 0.999$) and weight decay 10^{-5} . We apply linear warm-up over the first 5% of optimization steps, followed by cosine decay from $\eta_{\max} = 5 \times 10^{-4}$ to $\eta_{\text{final}} = 10^{-6}$. The objective combines MSE reconstruction loss with an auxiliary penalty on activations that never enter the top- k set, using $\lambda = 10^{-5}$.

979 C Computational Resources

980 This section reports the GPU resources used for training and experimentation. In total, the study
981 consumed approximately 202 GPU-hours on NVIDIA H100 and H200 hardware.

982 **RA-SAE.** Training required approximately 24 GPU-hours distributed across three NVIDIA H100
983 GPUs.

984 **Synthesized Images.** Generating the full D_{g_θ} (see Appendix A) required roughly 5 hours per
985 generator when distributed across four NVIDIA H200 GPUs. Across five generators, this amounted
986 to approximately 100 GPU-hours on NVIDIA H200 hardware.

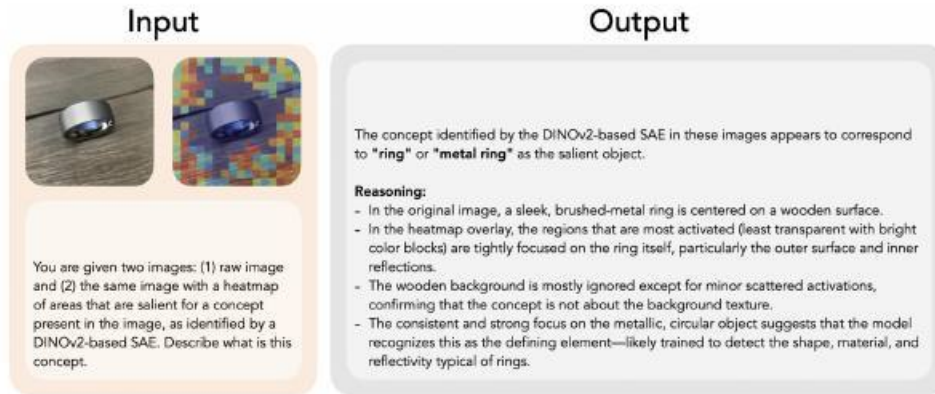
987 **● Distribution-Level Analysis.** Distribution-level energy-difference extraction required approxi-
988 mately 3 hours per generator on a single NVIDIA H200 GPU, for a total of ~ 15 GPU-hours.

989 **● Datapoint-Level Analysis.** Computing datapoint-level energy differences, ranking concepts
990 within each datapoint, and ranking datapoints by concept likewise required approximately 3 hours
991 per generator on one H200 GPU, totaling ~ 15 GPU-hours.

D Autointerpretability

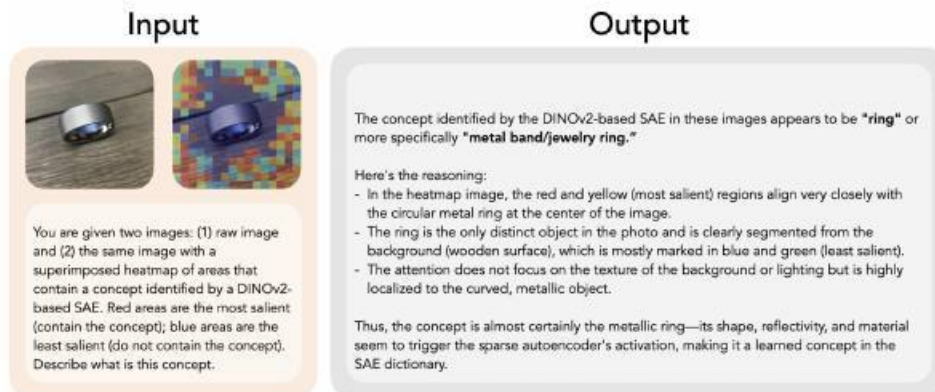
To automate the interpretation of concepts identified by the RA-SAE, we qualitatively evaluated whether Vision-Language Models (VLMs) could describe them. We queried ChatGPT 4o through the OpenAI API using several prompts and several formats of an image x'' in which the target concept was salient, and we recorded the descriptions produced by the VLM. This section summarizes the resulting observations. For evaluation, we permitted the VLM to provide its full reasoning; in a scalable deployment, the prompt could simply request only the final description, thereby filtering out reasoning text.

First, we supplied the VLM with two images: the raw x'' and a version of x'' overlaid with a heatmap intended to localize concept intensity:



As the example shows, the VLM did not correctly interpret the heatmap. Rather than describing the wood texture in the background, which is the actual salient region, it described the foreground ring, which received no attention from the RA-SAE.

To determine whether the failure was primarily a prompting issue, we repeated the same scenario with an expanded prompt that explained the heatmap in greater detail:



This prompt modification did not resolve the problem: the VLM continued to identify the ring as the prominent concept. We explored several additional prompting strategies, including prompts that described the heatmap more explicitly and even included the code used to generate it, but obtained similar results.

1011 We next provided only the heatmap-superimposed x'' , paired with a similar prompt explaining how
1012 the heatmap should be interpreted:

Input	Output
 You are given an image with a superimposed heatmap of areas that contain a concept identified by a DINOv2-based SAE. Red areas are the most salient (contain the concept); blue areas are the least salient (do not contain the concept). Describe what is this concept.	The concept identified by the DINOv2-based SAE in this image appears to be "ring" or more broadly, "circular metallic object". The most salient (red) areas are focused tightly around the ring itself—especially along its edges and reflective interior—indicating that the model has strongly associated those regions with the concept. The surrounding wooden surface is mostly in blue and green, which shows it is not relevant to the concept, supporting the idea that the model is isolating the ring as the key object of interest.

1013 Again, the VLM failed to identify the wooden background as the prominent concept and instead
1014 focused on the ring. As before, additional prompting strategies using this image format produced
1015 largely unchanged outcomes.

1016 These observations suggest that the heatmap alone is insufficient to direct the VLM toward the salient
1017 regions, partly because less relevant image regions appear to distract the model. To address this issue,
1018 we replaced the heatmap-superimposed x'' with an alpha-masked version of x'' :

Input	Output
 You are given an image where only areas that contain a concept identified by a DINOv2-based SAE are preserved; areas without the concept were removed. Describe what is this concept.	The concept identified and preserved in this image appears to be "wood texture" or "wood grain." The retained areas show consistent patterns typical of wooden surfaces—linear, fibrous streaks in earthy tones like beige, gray, and brown. The selective masking highlights parts of the image where this natural texture is most prominent, suggesting the SAE learned to isolate the visual features of wood (likely flooring or a table).

1019 With this representation, the VLM correctly identified wood texture as the dominant concept. We
1020 observed similarly reliable behavior across a range of concepts and images.

1021 The image formatting and prompt shown above produced the most reliable results in our qualitative
1022 evaluation. Nevertheless, the evaluation remains limited by its qualitative character, the absence of
1023 ground-truth annotations, and its focus on a single VLM. Future work on SAE concept autointer-
1024 pretability can extend this analysis with broader model coverage and more systematic validation.

E Additional Results: Model-Specific Blindspots

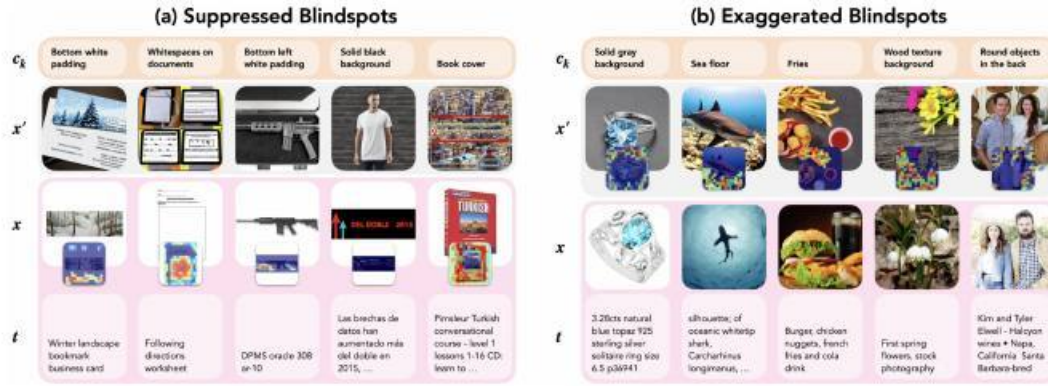


Figure 14: Examples of conceptual blindspots in **Stable Diffusion 1.5**. For each concept, we show the prototypical natural image for suppressed blindspots or the prototypical synthesized image for exaggerated blindspots, selected by highest absolute activation. The concept-specific spatial heatmap is superimposed on the image.

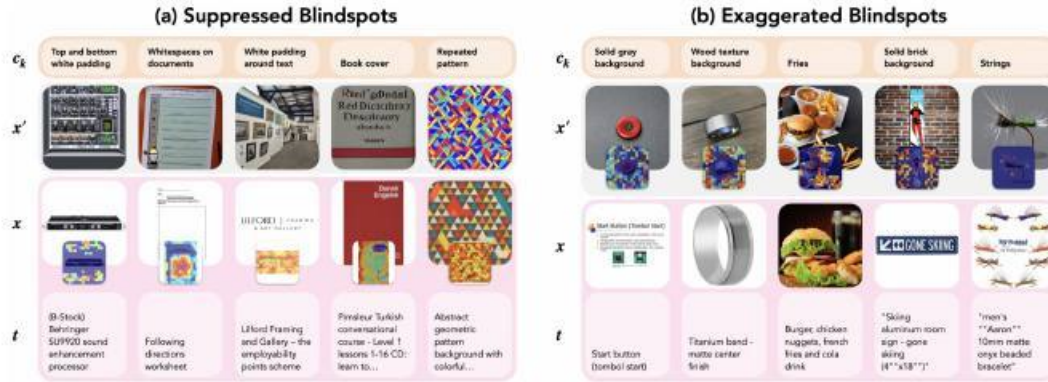


Figure 15: Examples of conceptual blindspots in **Stable Diffusion 2.1**. For each concept, we show the prototypical natural image for suppressed blindspots or the prototypical synthesized image for exaggerated blindspots, selected by highest absolute activation. The concept-specific spatial heatmap is superimposed on the image.

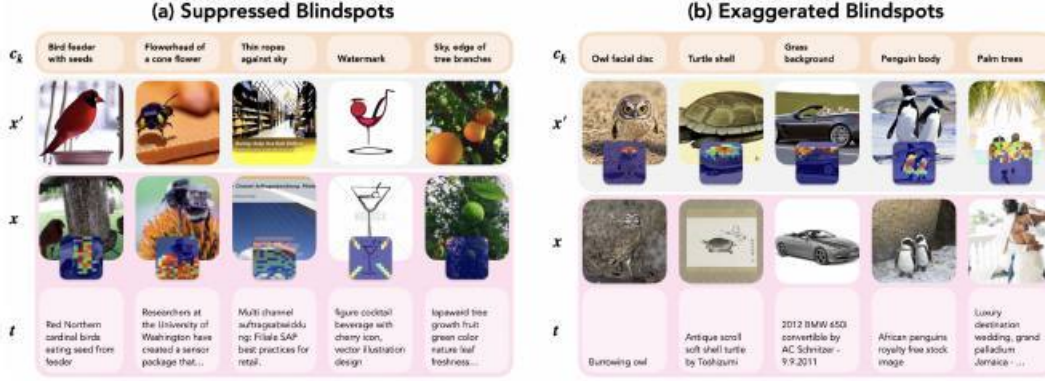


Figure 16: Examples of conceptual blindspots in **Kandinsky**. For each concept, we show the prototypical natural image for suppressed blindspots or the prototypical synthesized image for exaggerated blindspots, selected by highest absolute activation. The concept-specific spatial heatmap is superimposed on the image.



Figure 17: Examples of conceptual blindspots in **PixArt**. For each concept, we show the prototypical natural image for suppressed blindspots or the prototypical synthesized image for exaggerated blindspots, selected by highest absolute activation. The concept-specific spatial heatmap is superimposed on the image.

F Additional Results: Higher-order Blindspots with Compositional Discrepancy

Thus far, the analysis has focused on individual concept activations. Visual scenes, however, are rarely constituted by isolated concepts; rather, they are organized through rich co-occurrence patterns that encode compositional semantics. We therefore examine whether generative models preserve this higher-order structure by analyzing the co-activation matrix $Z^T Z$, which captures pairwise correlations in concept usage.

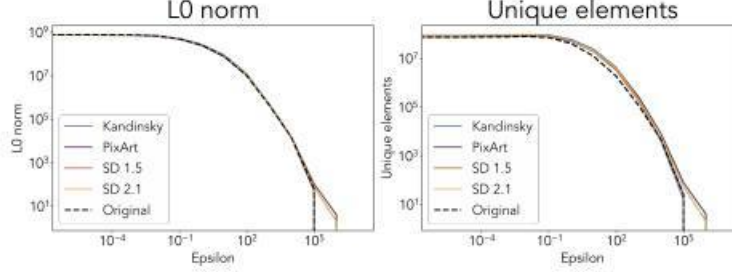


Figure 18: **Sparsity and Structural Divergence.** Left: L_0 norm of the co-occurrence matrix ZZ^T as a function of the threshold ϵ , indicating how many entries remain active in each model. Right: number of unique entries in the synthesized distribution relative to the natural distribution. All evaluated models preserve global sparsity structure while diverging in activation content.

Somewhat surprisingly, when evaluated at the level of binary structure, diffusion models approximate the global sparsity of the natural co-occurrence matrix with high fidelity. As Fig. 18 (left) shows, the ℓ_0 norm of $Z^T Z$, thresholded at varying ϵ values, closely tracks between natural and synthesized distributions for all models. This indicates that the coarse connectivity of the conceptual graph, namely which concepts co-activate at all, is largely preserved. Formally, $Z^T Z$ can be interpreted as the adjacency matrix of a weighted, undirected graph over concepts, with edge weights corresponding to co-activation strength across the dataset.

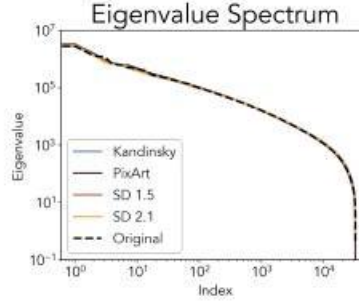


Figure 19: **Spectral Structure of Co-occurrence.** Log-log eigenvalue spectra for co-occurrence matrices ZZ^T across the evaluated models and the natural distribution. All evaluated models reproduce the heavy-tailed decay observed in the natural distribution.

However, as Fig. 18 (right) illustrates, the specific content of these co-activations diverges: a substantial fraction of entries in the model-generated $Z^T Z$ are not shared with the natural baseline. This suggests that, although the capacity for compositionality is retained, the identities of active pairings can shift, potentially reflecting model-specific inductive biases or training artifacts. To probe the internal organization of these co-occurrence patterns, we next turn to spectral analysis. Fig. 19 reports the eigenvalue spectra of the co-occurrence matrices for each model and for the natural distribution. All spectra display heavy-tailed decay consistent with power-law behavior, indicating that generative models preserve the overall rank structure and variance allocation across conceptual dimensions.

We further examine dominant-direction alignment through cosine-similarity heatmaps between the top 100 eigenvectors of synthesized and natural co-occurrence matrices (Fig. 20). Although all models ex-

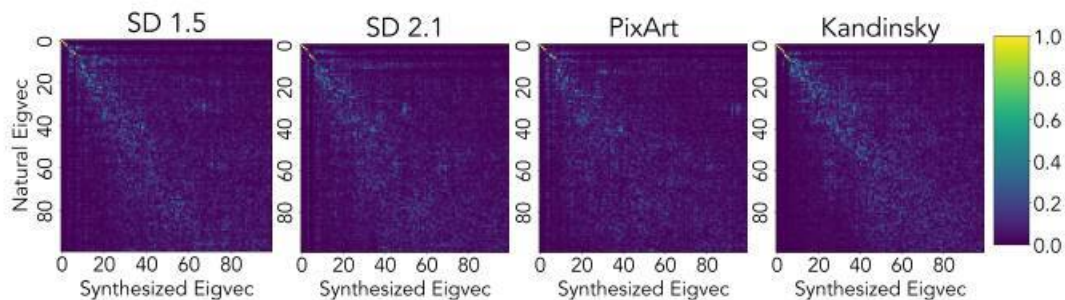


Figure 20: ● **Concept Basis Similarity.** Cosine-similarity heatmaps compare the top 100 eigenvectors of natural and synthesized co-occurrence matrices ZZ^T . Diagonal structure indicates alignment among dominant conceptual directions, with the degree of alignment varying across the four models.

hibit partial diagonal alignment, implying overlap among principal concept axes, off-diagonal entries reveal rotations and higher-mode mismatches that indicate deviations in compositional geometry.

Collectively, these findings show that diffusion models approximate the global form of concept co-activation surprisingly well, yet exhibit subtle and structured deviations under spectral inspection. Such higher-order discrepancies may contribute to failures in generating coherent multi-object scenes or relational concepts.

1058 G Additional Results: Blindspot Stress Testing

1059 To stress-test the blindspots identified by our method, we collected a range of prompts describing
 1060 each blindspot and used them to generate multiple images. We then compared outputs from models
 1061 for which the concept was identified as a blindspot with outputs from models for which it was not.

1062 Specifically, ChatGPT-4o was prompted as follows: *I want to generate an image of the following*
 1063 *concept: "<blindspot>". Suggest 50 prompts highlighting this concept to be used as input for a*
 1064 *text-to-image model. Return these as a list of strings in Python.* For each prompt, we generated
 1065 five images and analyzed them with our custom RA-SAE model (see Appendix B), which ranked
 1066 them according to the intensity of the desired concept. All images were then manually reviewed to
 1067 determine whether the blindspot was successfully depicted.

1068 As the following suppressed-concept examples show, partial aspects of the target concept sometimes
 1069 appeared, such as a holder or string for the bird feeder blindspot or a circular hole for the glossy
 1070 DVD disc blindspot; nevertheless, the models generally failed to synthesize the complete concept.
 1071 This behavior is consistent with our method’s assessment and supports the stress test’s validity.

1072 G.1 Bird Feeder Blindspot in Kandinsky



Figure 21: Examples of images generated from prompts involving the bird feeder concept during stress testing. For Kandinsky, our method identifies this concept as a suppressed conceptual blindspot, matching the observed behavior that the model cannot generate a corresponding image. By contrast, SD 1.5, for which this concept is not identified as a blindspot, successfully generates the concept.



Figure 22: Examples of images generated from prompts involving the glossy DVD disc concept during stress testing. For SD 1.5, our method identifies this concept as a suppressed conceptual blindspot, matching the observed behavior that the model cannot generate a corresponding image. By contrast, Kandinsky, for which this concept is not identified as a blindspot, successfully generates the concept.

1074 **H Additional Examples: Synthesized Images**



Figure 23: Additional LAION-5B image-caption examples, together with matching images generated from the same prompt by SD 1.5, SD 2.1, PixArt, and Kandinsky.

1075 I Concentration bounds for δ

1076 In our experiments, we estimate $\delta(k)$ using $n = 10,000$ paired samples for each concept. While this
 1077 budget is modest, it raises the natural question of whether it suffices to obtain reliable estimates. To
 1078 address this, we derive a concentration bound on the empirical estimator $\hat{\delta}_n(k)$ using McDiarmid’s
 1079 inequality [142]. The resulting bound is tight and demonstrates that even with relatively few samples,
 1080 we can obtain fast and accurate estimates of concept bias.

1081 **Theorem 4** (Concentration of $\hat{\delta}_n(k)$). *We assume that the concept score $\xi_k(\mathbf{x})$ takes values in $[a, b]$
 1082 almost surely for all images $\mathbf{x}$ drawn from either $\mathcal{D}_\mathcal{X}$ or $\mathcal{D}'_\mathcal{X}$. Let n paired samples $(\mathbf{x}_i, \mathbf{x}'_i)_{i=1}^n$ be
 1083 drawn independently with $\mathbf{x}_i \sim \mathcal{D}_\mathcal{X}$ and $\mathbf{x}'_i \sim \mathcal{D}'_\mathcal{X}$, and define the empirical estimator*

$$\hat{\delta}_n(k) := \sigma \left(\frac{1}{n} \sum_{i=1}^n \xi_k(\mathbf{x}_i) - \frac{1}{n} \sum_{i=1}^n \xi_k(\mathbf{x}'_i) \right).$$

1084 Let $M := b - a$ and $L := M/4$. Then for every $\varepsilon > 0$, the deviation satisfies

$$\mathbb{P} \left(\left| \hat{\delta}_n(k) - \delta(k) \right| > \varepsilon \right) \leq 2 \exp \left(-\frac{2n\varepsilon^2}{L^2} \right).$$

1085 *Proof.* The function $x \mapsto \sigma(x)$ is $1/4$ -Lipschitz, since $|\sigma'(x)| \leq 1/4$ for all x .

1086 Viewing $\hat{\delta}_n(k)$ as a function of the $2n$ independent variables $(\mathbf{x}_1, \dots, \mathbf{x}_n, \mathbf{x}'_1, \dots, \mathbf{x}'_n)$, changing a
 1087 single argument alters the inner difference of means by at most M/n , and the outer sigmoid scales this
 1088 by at most $1/4$. Hence, the bounded difference constant for each coordinate is $(M/n)(1/4) = L/n$.

1089 By McDiarmid’s inequality [142],

$$\mathbb{P} \left(\left| \hat{\delta}_n(k) - \delta(k) \right| > \varepsilon \right) \leq 2 \exp \left(-\frac{2\varepsilon^2}{\sum_{j=1}^{2n} (L/n)^2} \right) = 2 \exp \left(-\frac{2n\varepsilon^2}{L^2} \right),$$

1090 which proves the claim. □

1091 In practice, most concept scores $\delta(k)$ are sparse, with the majority concentrated near zero and only
 1092 a small number reaching values as high as 10. The concentration bound implies that, even for the
 1093 largest observed biases, a sample size of $n = 10,000$ produces estimates of $\hat{\delta}_n(k)$ that deviate from
 1094 the true value by at most a small ε with high probability. This result justifies our sampling strategy
 1095 and confirms that accurate bias measurements are attainable from a limited evaluation set.

1096 J Monotonicity and Calibration-Free Interpretation of δ

1097 When analyzing blindspots, our objective is to rank concepts by the severity of their generative
 1098 bias, and we use $\delta(k)$ for this purpose. A natural concern is whether this score distorts more direct
 1099 quantities such as the energy gap or odds ratio. The following result establishes that $\delta(k)$ is a strictly
 1100 increasing reparameterization of both quantities and therefore preserves their ordering, which means
 1101 that no additional calibration is required when ranking concepts by $\delta(k)$.

1102 **Theorem 5** (Monotonicity and Calibration of $\delta_{g_\theta \leftrightarrow G}$). *For every concept index k define the energy*
 1103 *gap*

$$\Delta_k = \mathbb{E}_{\mathbf{x}' \sim \mathcal{D}'_X}[\xi_k(\mathbf{x}')] - \mathbb{E}_{\mathbf{x} \sim \mathcal{D}_X}[\xi_k(\mathbf{x})],$$

1104 *the associated odds ratio $\rho_k = \exp(\Delta_k)$, and the energy-difference score*

$$\delta(k) = \frac{1}{1 + \exp(-\Delta_k)} = \frac{\rho_k}{1 + \rho_k}.$$

1105 *Then $\delta(k)$ is a strictly increasing bijection of both Δ_k and ρ_k , so ranking concepts by any one of*
 1106 *$\delta(k)$, Δ_k , or ρ_k produces exactly the same ordering.*

1107 *Proof.* The logistic sigmoid satisfies $\sigma'(x) = \sigma(x)(1 - \sigma(x)) > 0, \forall x \in \mathbb{R}$; hence σ and therefore
 1108 $\delta(k) = \sigma(\Delta_k)$ grow strictly with Δ_k . Because the exponential map is also strictly increasing and
 1109 bijective $\mathbb{R} \rightarrow (0, \infty)$, setting $\rho_k = \exp(\Delta_k)$ preserves order and gives $\Delta_k = \log \rho_k$. Substituting
 1110 this identity into σ yields $\delta(k) = \sigma(\log \rho_k) = \rho_k / (1 + \rho_k)$, which is the composition of two
 1111 strictly increasing bijections and is therefore itself strictly increasing and bijective in ρ_k . Since strict
 1112 monotonic functions never reverse inequalities, the three quantities share the same total order over
 1113 concepts.

1114

□

1115 Consequently, ranking concepts by $\delta(\cdot)$ is strictly equivalent to ranking them by energy gap or by
 1116 conceptual generation odds ρ_k . No calibration is required, because all three quantities induce the
 1117 same total ordering over concepts.

K Stability of FID Under SAE Embeddings

In this section, we establish a quantitative relationship between Fréchet Inception Distance (FID) computed in the original d -dimensional activation space and FID computed after applying a potentially overcomplete SAE dictionary $D \in \mathbb{R}^{k \times d}$ with $k \gg d$. Throughout, we assume that D has orthonormal columns but need not be square, i.e.

$$D^\top D = I_d, \quad \text{while} \quad DD^\top \neq I_k.$$

We begin with the elementary square case: if D is not overcomplete, is orthogonal, and satisfies $k = d$, then the map from A to Z is an isometry, and FID is therefore exactly preserved. Because this case is not representative of the overcomplete dictionaries used in practice, we then consider the overcomplete setting and show that FID distortion can be bounded by the extremal singular values of D . We work with the Wasserstein-2 metric $\mathcal{W}_2$, noting that FID is precisely $\mathcal{W}_2^2$ specialized to Gaussian distributions.

For a probability measure μ on $\mathbb{R}^d$, we write $D_\# \mu$ for its push-forward under D , i.e. $D_\# \mu(z) = \mu(D^{-1}z)$. Let $\sigma_{\min}$ and $\sigma_{\max}$ denote the minimal and maximal singular values of D , equivalently the square roots of the extremal eigenvalues of $DD^\top$:

$$\sigma_{\min}^2 I_k \preceq DD^\top \preceq \sigma_{\max}^2 I_k.$$

Empirically, one often observes $\sigma_{\min}, \sigma_{\max} \approx 1$, although the proof does not rely on this approximation. We start with a simple lemma for the non-overcomplete case.

Lemma 1 (Isometry under exact orthogonality). *Suppose $k = d$ and $D^\top D = DD^\top = I_d$. Then D is an isometry: $\|Dv\|_2 = \|v\|_2$ for all $v \in \mathbb{R}^d$. Consequently, for any probability measures μ, ν on $\mathbb{R}^d$ with finite second moment,*

$$\mathcal{W}_2(D_\# \mu, D_\# \nu) = \mathcal{W}_2(\mu, \nu).$$

Proof. Orthogonality of D preserves the Euclidean norm, and the push-forward commutes with the map inside the $\mathcal{W}_2$ infimum; because the integrand is unchanged, the infimum has the same value. $\square$

This square case is unrealistic for SAEs, which typically rely on overcompleteness to extract meaningful and interpretable concepts. In the overcomplete case, D is no longer orthonormal as a square matrix, but it may still define a column-orthonormal dictionary. We use this property to obtain a bound governed by the extremal singular values of $D^\top D$.

Theorem 6 (FID under column orthogonal embeddings). *Let $D \in \mathbb{R}^{k \times d}$ satisfy $D^\top D = I_d$ and denote by $0 < \sigma_{\min} \leq \sigma_{\max}$ the extreme singular values of $DD^\top$. Given two data matrices $A, A' \in \mathbb{R}^{n \times d}$ (rows are sample vectors), set $Z = AD^\top \in \mathbb{R}^{n \times k}$ and $Z' = A'D^\top \in \mathbb{R}^{n \times k}$. Then*

$$\sigma_{\min}^2 \text{FID}(A, A') \leq \text{FID}(Z, Z') \leq \sigma_{\max}^2 \text{FID}(A, A').$$

Proof. Write μ for the empirical measure of A and ν for that of A' , i.e.

$$\mu = \frac{1}{n} \sum_{i=1}^n \delta_{A_{i,:}}, \quad \nu = \frac{1}{n} \sum_{i=1}^n \delta_{A'_{i,:}}.$$

For any coupling $\pi \in \Pi(\mu, \nu)$ (i.e. a probability measure on $\mathbb{R}^d \times \mathbb{R}^d$ with marginals μ, ν) we have, by the extremal singular value bound,

$$\sigma_{\min}^2 \|x - y\|_2^2 \leq \|D(x - y)\|_2^2 \leq \sigma_{\max}^2 \|x - y\|_2^2, \quad \forall (x, y) \in \mathbb{R}^d \times \mathbb{R}^d.$$

Integrating with respect to an arbitrary coupling $\pi \in \Pi(\mu, \nu)$ yields

$$\sigma_{\min}^2 \int \|x - y\|_2^2 d\pi \leq \int \|D(x - y)\|_2^2 d\pi \leq \sigma_{\max}^2 \int \|x - y\|_2^2 d\pi.$$

The middle integral is precisely the transport cost of the pushed-forward coupling $(D \times D)_\# \pi$ between $\mu_D := D_\# \mu$ and $\nu_D := D_\# \nu$. Because the inequalities hold for every π , they also hold

1153 for the optimal couplings attaining $\mathcal{W}_2(\boldsymbol{\mu}, \boldsymbol{\nu})$ and $\mathcal{W}_2(\boldsymbol{\mu}_D, \boldsymbol{\nu}_D)$, although these two optima need not
 1154 coincide. Taking the infimum over $\boldsymbol{\pi}$ term-wise makes this relationship explicit:

$$\sigma_{\min}^2 \inf_{\boldsymbol{\pi} \in \Pi(\boldsymbol{\mu}, \boldsymbol{\nu})} \int \|\mathbf{x} - \mathbf{y}\|_2^2 d\boldsymbol{\pi} \leq \inf_{\boldsymbol{\pi} \in \Pi(\boldsymbol{\mu}, \boldsymbol{\nu})} \int \|D(\mathbf{x} - \mathbf{y})\|_2^2 d\boldsymbol{\pi} \leq \sigma_{\max}^2 \inf_{\boldsymbol{\pi} \in \Pi(\boldsymbol{\mu}, \boldsymbol{\nu})} \int \|\mathbf{x} - \mathbf{y}\|_2^2 d\boldsymbol{\pi}.$$

1155 Hence

$$\sigma_{\min}^2 \mathcal{W}_2^2(\boldsymbol{\mu}, \boldsymbol{\nu}) \leq \mathcal{W}_2^2(\boldsymbol{\mu}_D, \boldsymbol{\nu}_D) \leq \sigma_{\max}^2 \mathcal{W}_2^2(\boldsymbol{\mu}, \boldsymbol{\nu}).$$

1156 Recognizing $\text{FID}(\cdot, \cdot) = \mathcal{W}_2^2(\cdot, \cdot)$ for the Gaussian surrogate and substituting $(\mathbf{A}, \mathbf{A}')$ (respectively,
 1157 $(\mathbf{Z}, \mathbf{Z}')$) completes the proof. $\square$

1158 Theorem 6 therefore shows that applying a column-orthogonal overcomplete SAE dictionary cannot
 1159 distort Fréchet Inception Distance by more than the square of its extremal singular values. When
 1160 $\mathbf{D}$ is nearly orthogonal—as we empirically observe through $\sigma_{\min}, \sigma_{\max} \approx 1$ —FID measured in the
 1161 SAE feature space remains close to canonical FID. $\square$

1162